

CS1 - Introduction to Programming in Java

Assignment 1

(Chapter 5, Programming Challenge 8. Conversion Program)

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8. Conversion Program

Write a program that asks the user to enter a distance in meters. The program will then present the following menu of selections:

1. Convert to kilometers
2. Convert to inches
3. Convert to feet
4. Quit the program

The program will convert the distance to kilometers, inches, or feet, depending on the user's selection. Here are the specific requirements:

- Write a void method named `showKilometers`, which accepts the number of meters as an argument. The method should display the argument converted to kilometers.

Convert the meters to kilometers using the following formula:

```
kilometers = meters * 0.001
```

- Write a void method named `showInches`, which accepts the number of meters as an argument. The method should display the argument converted to inches. Convert the meters to inches using the following formula:

```
inches = meters * 39.37
```

- Write a void method named `showFeet`, which accepts the number of meters as an argument. The method should display the argument converted to feet. Convert the meters to feet using the following formula:

```
feet = meters * 3.281
```

- Write a `void` method named `menu` that displays the menu of selections. This method should not accept any arguments.
- The program should continue to display the menu until the user enters 4 to quit the program.
- The program should not accept negative numbers for the distance in meters.
- If the user selects an invalid choice from the menu, the program should display an error message.

Sample Output

```
Enter a distance in meters: 500 [Enter]
1. Convert to kilometers
2. Convert to inches
3. Convert to feet
4. Quit the program
Enter your choice: 1 [Enter]
500 meters is 0.5 kilometers.
1. Convert to kilometers
2. Convert to inches
3. Convert to feet
4. Quit the program
Enter your choice: 3 [Enter]
500 meters is 1640.5 feet.
1. Convert to kilometers
2. Convert to inches
3. Convert to feet
4. Quit the program
Enter your choice: 4 [Enter]
Bye!
```

Note:

[1] In the sample output shown here, the option 2 is missing.

Make sure that your testing includes all 4 choices.

[2] Ensure that you display a valid error message if the users makes an incorrect choice.

[3] Ensure that the program does not accept the negative numbers for the distance in meters.

[4] Make sure that the program continues to display the menu option till the user choses the option 4

Testing and sample output

```
Enter a distance in meters: 500 [Enter]
```

1. Convert to kilometers
2. Convert to inches
3. Convert to feet
4. Quit the program

```
Enter your choice: 1 [Enter]
```

```
500 meters is 0.5 kilometers.
```

1. Convert to kilometers
2. Convert to inches
3. Convert to feet
4. Quit the program

```
Enter your choice: 3 [Enter]
```

```
500 meters is 1640.5 feet.
```

1. Convert to kilometers
2. Convert to inches
3. Convert to feet
4. Quit the program

```
Enter your choice: 4 [Enter]
```

```
Bye!
```

Select Option 2 as additional test case.

5
Error: Invalid Choice. Try again!

1. Convert to kilometers
 2. Convert to inches
 3. Convert to feet
 4. Quit the program
- Enter your choice:

What to submit?

Submit these two files separately. Do not submit the zip files. And do not submit the whole project folder. I just need these two files – java file and output text file.

- A1_Conversion.java
- A1_Conversion_output.txt

Evaluation:

1. showKilometers method is implemented
2. showInches method is implemented
3. showFeet method is implemented
4. menu method is implemented
5. main() method is implemented meeting the requirements.